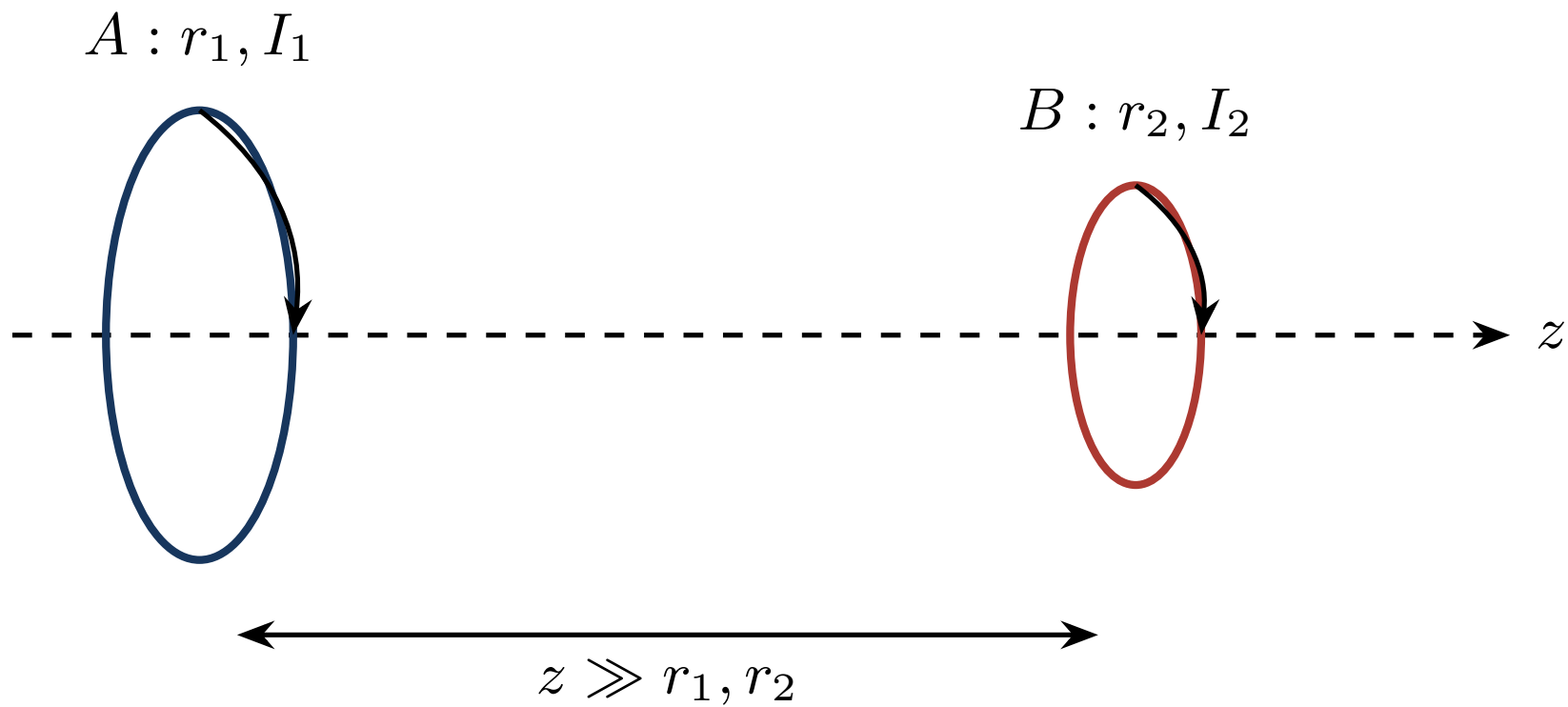
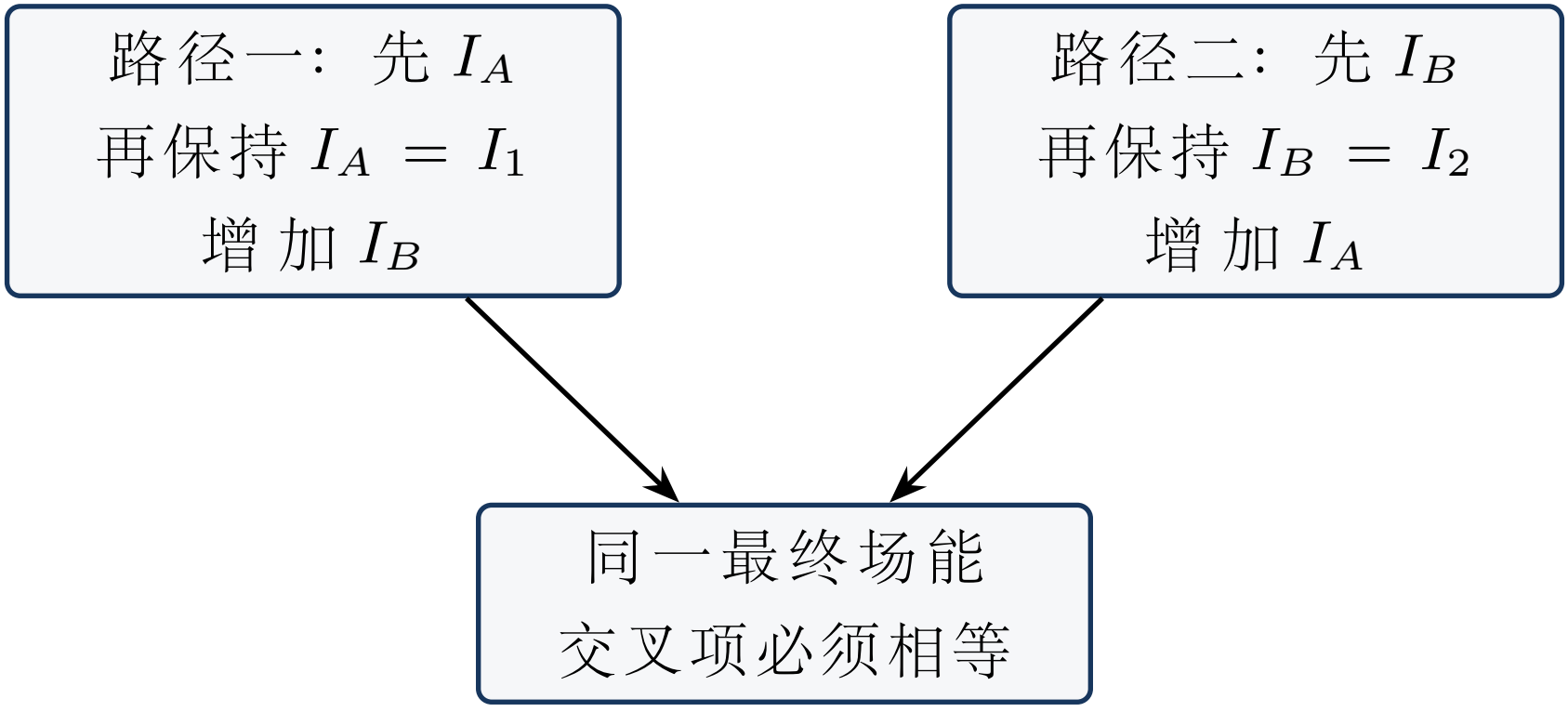
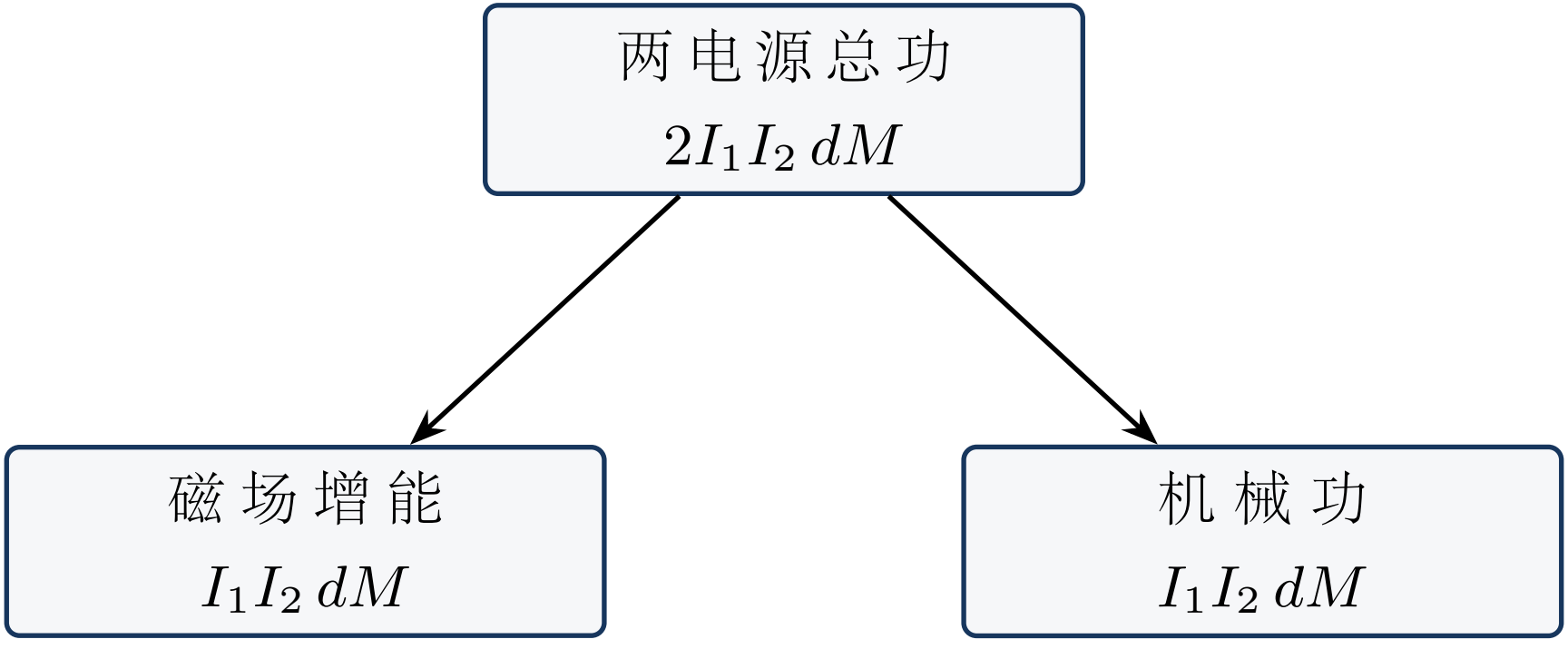


先证明固定线性无耗散介质中 $M_{AB} = M_{BA}$

再用远轴磁偶极近似求力

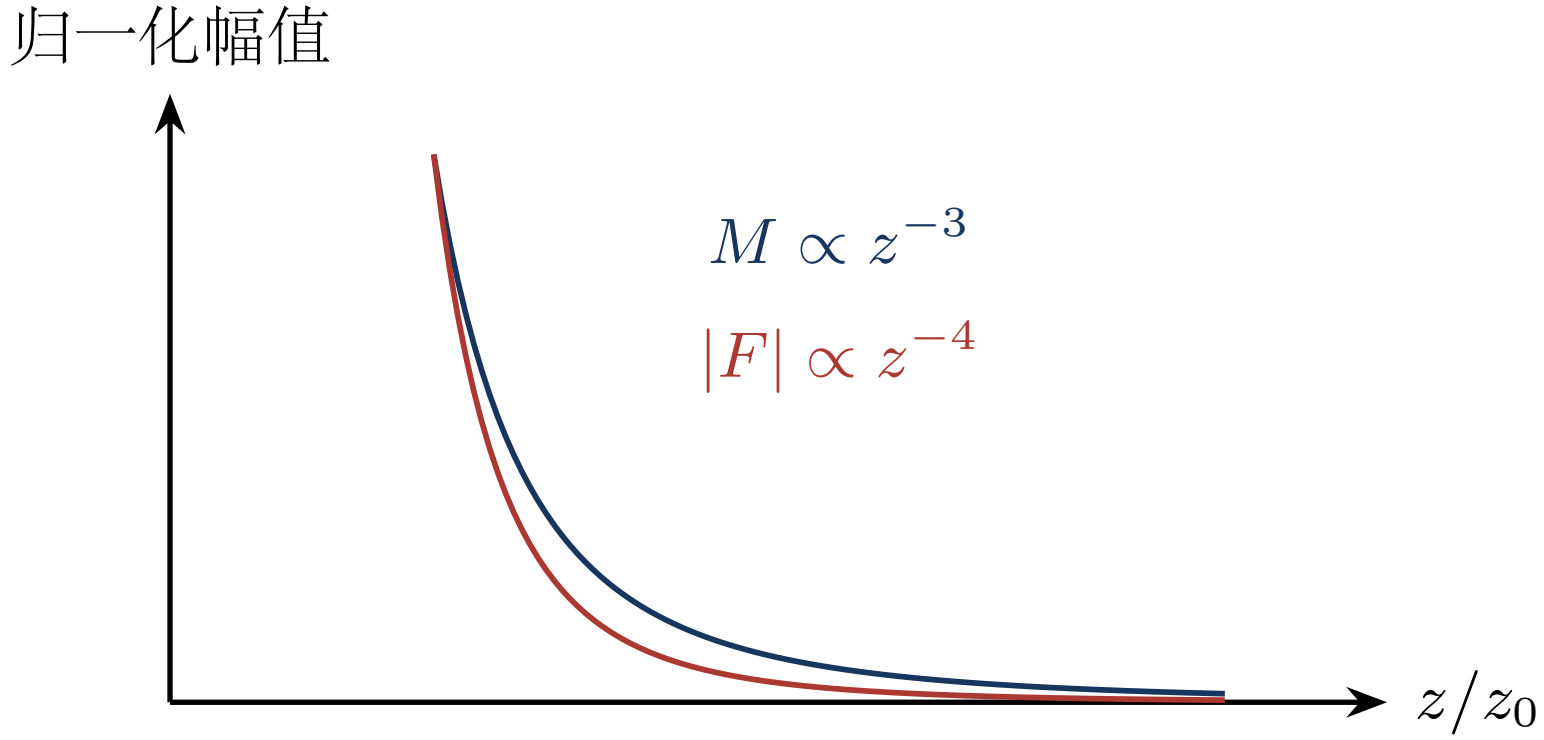


$$W_1 = W_0 + M_{AB} I_1 I_2;\ W_2 = W_0 + M_{BA} I_1 I_2$$

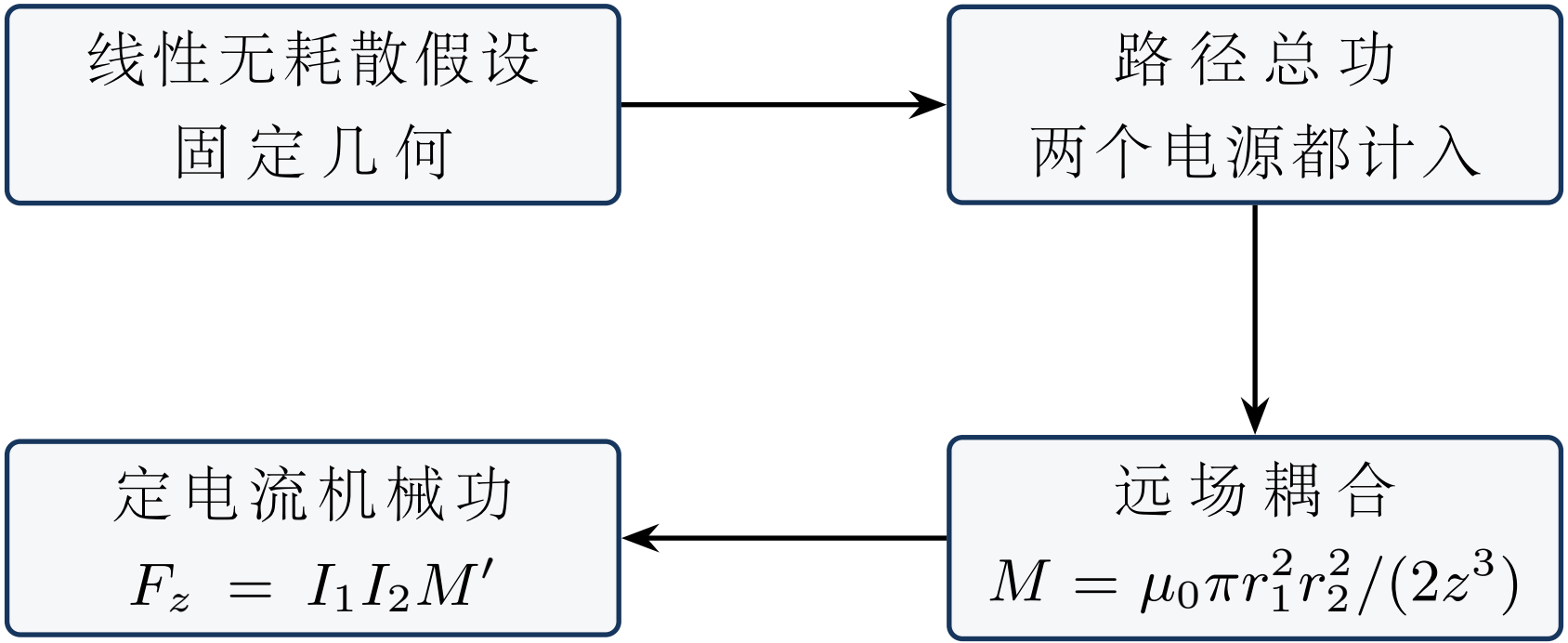


保持两电流需要电源持续交换能量

$$F_z = I_1I_2M'(z)$$



同向电流：$M > 0$，$M'(z) < 0$，沿减小间距方向吸引



非互易或有耗散介质不直接套用可逆路径证明